

RVEM

Reliability & Validation (Explainability) Module

EVA Machine Learning Reasoning Layer — Module Specification

Version 1.0 | References: EVA-ML-Master.docx

1. Purpose

The Reliability & Validation Module verifies that the selected model behaves in a logically consistent and explainable manner relative to the dataset and previously established analytical reasoning. The goal is not only to know that the model predicts correctly — but to understand why it predicts correctly.

Core Rule (from Master PRD)

EVA never trusts a model more than it trusts the evidence already observed in the dataset. This module is where that rule is enforced. If a model contradicts strong prior evidence, it must be flagged — not deployed.

2. Design Philosophy

A model that cannot be explained cannot be trusted. A model may achieve high predictive performance while relying on irrelevant variables, indirect leakage, or unstable correlations. Such models produce unreliable real-world decisions. The model is treated as a hypothesis tester — not an authority.

3. Inputs from Global Analysis Ledger

- `model_evaluation_record` — selected model and evaluation results
- `hypotheses` — prior findings from Investigation & Hypothesis Engine
- `feature_registry` — feature reasoning and domain context
- `dataset_identity` — domain and row meaning

4. Feature Influence Analysis

The module determines which variables most influenced the model's predictions. It identifies important features, the direction of their influence, and the magnitude of each contribution. This establishes the factual basis for all downstream explainability.

5. Hypothesis Consistency Check

The module compares model behavior with earlier analytical findings from the GAL hypotheses key. For each major hypothesis, the module checks whether the model supports or contradicts it.

A contradiction does not automatically invalidate the model. But every contradiction must be explained and documented. If the contradiction cannot be explained, the model is rejected.

Example: If earlier analysis found that inactivity increases churn, the model should rely on inactivity-related features. If it ignores them in favor of an unrelated proxy, that is a reasoning failure requiring investigation.

6. Leakage & Proxy Detection

The module inspects whether the model is indirectly using forbidden information. This includes encoded identifiers, near-duplicate predictors, post-outcome variables, and proxies strongly tied to the target. If leakage is detected, the model is immediately rejected and the pipeline is sent back to the PFTDC for Learning View revision.

7. Sensitivity Testing

The module tests model behavior under small controlled input changes. It evaluates whether predictions change reasonably and whether small irrelevant changes cause large unexpected prediction shifts. Unstable behavior indicates unreliable reasoning and triggers rejection.

8. Case-Level Explanation

For selected records, the module generates case-level explanations including: why a prediction was made, which features contributed most, and how strong each influence was. These explanations are used by the Dashboard Composer and Report Generator.

9. Confidence Assessment

The module evaluates prediction confidence reliability. Not all predictions should be treated equally. The module identifies high-confidence predictions, uncertain predictions, and borderline cases. Uncertain predictions are flagged for caution in all downstream outputs.

10. Validation Status

The module assigns one of three statuses to the selected model:

Status	Meaning & Downstream Behavior
Validated	Model reasoning fully aligns with evidence. Deployed normally.
Conditionally Validated	Model is usable but requires caution. Dashboard displays persistent caution banner. Monitoring runs at 2x standard frequency. Report must include a dedicated limitations section. Human acknowledgement required for high-stakes use cases.
Rejected	Reasoning is unreliable, inconsistent, or leakage was detected. Model cannot be deployed. Pipeline halts. MCG and TEM are notified for revision.

11. Output: model_validation_record

Field	Contents
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Feature Influence Summary	What drove predictions and with what strength
Hypothesis Alignment	Agreement or contradiction with analytical findings
Leakage Inspection	Whether improper signals were detected
Sensitivity Behavior	Stability of predictions under controlled variation
Prediction Confidence Profile	Distribution of confidence levels
Validation Status	Validated / Conditionally Validated / Rejected
Contradiction Log	All hypothesis contradictions found and their explanations

12. Operational Rules

1. A model cannot be deployed without passing this module.
2. All hypothesis contradictions must be documented. Unexplained contradictions result in rejection.
3. Leakage detection results in immediate rejection and PFTDC restart.
4. All predictions must be explainable at the case level.
5. Confidence must be assessed and attached to the model_validation_record.
6. Conditionally Validated models must trigger all specified downstream behaviors — dashboard warning, increased monitoring frequency, and report limitations section.
7. The core rule from the Master PRD applies here: the model is never trusted more than the prior evidence.

13. Relationship to Other Modules

Module	Relationship
TEM	Provides the selected model for validation
PFTDC	Receives leakage rejection and restarts Learning View construction
PMDD	Deploys only Validated or Conditionally Validated models
Dashboard Composer	Receives case-level explanations and validation status
Report Generator	Communicates reasoning, limitations, and status
Master PRD	Defines conditional validation operational behavior